C.A.N.I.S.

Containment Analysis & Nuclear Intelligence System

DRUM STORE INSPECTION REPORT

CANIS-112205-adaptive_aisle-20f6
Drum Store 1 (simulated)
start time not logged

VERDICT

1 drum needs action and 1 needs a second check. 4 showed no problem.

1 aisle section restricted; 1 not checked. Aisle ratings are simulation flags; site staff decide access.



Action needed 1 Check again 1 No problem found 4

DRUMS CHECKED	ACTION NEEDED	CHECK AGAIN	NO PROBLEM FOUND
5 / 6 verdict reached	1 red	1 amber	4 green
PEAK GAMMA	**AISLE SECTIONS**	**ROBOT DOSE**	**MISSION TIME**
1.71 µSv/h at 00:59 · restricted above 7.5	2 1 restricted · 1 not checked	0.0283 µSv no limit logged	1 min 34 s simulated time

WHAT TO DO NOW — MOST URGENT FIRST

	DRUM	RESULT	SEEN	DO THIS
1	**A-L-01-T2**	ACTION NEEDED	Rust	Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).
2	**A-L-01-T3**	CHECK AGAIN	Nothing found	Read the drum label in person and confirm which drum this is. Then repeat the inspection.

THIS INSPECTION

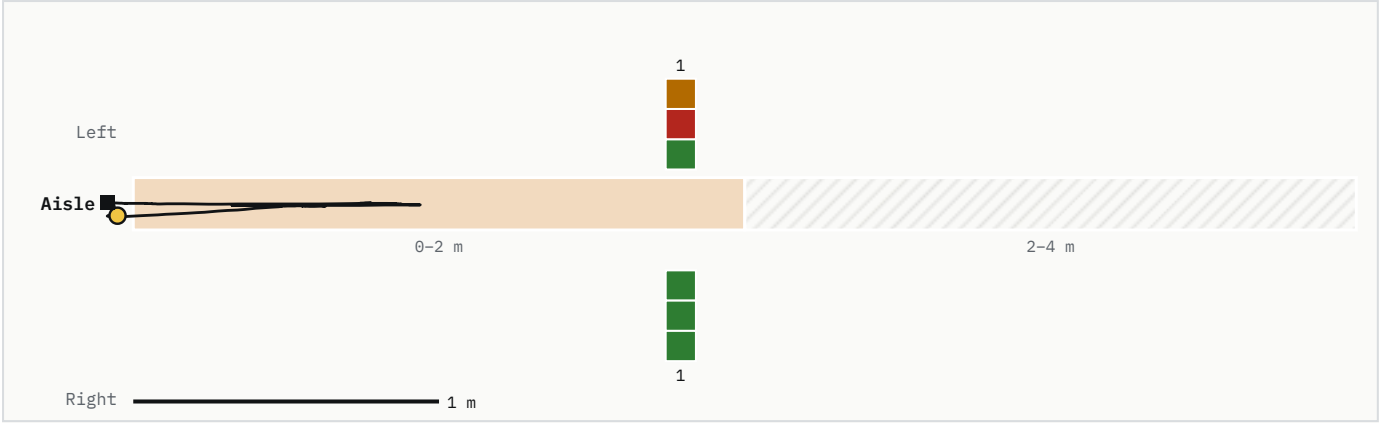
Store	Drum Store 1 (simulated)	Report	CANIS-112205-adaptive_aisle-20f6
Started	not logged	Mission time	1 min 34 s
Inspected by	C.A.N.I.S. robot (Spot)	Drums in scope	6
Report software	5b032b6	Status	Practice run, simulated store

Not measured on this round: drum surface temperature, surface wipes (swabs), waste category (LLW / ILW). Tables print “not measured”; no value is estimated.

01 Aisle survey

2 SECTIONS · 2 M EACH

Each aisle is split into 2-metre sections measured from where the robot started, and each section takes the worst rating that applies. A section is only rated low gamma if radiation was measured there.



Keep out · Restricted · Low gamma · Not checked · Not walked · Drum: action needed · Drum: check again · Drum: no problem found · Start · End · Robot's route

Racks are drawn folded out flat: the shelf nearest the aisle is the bottom one. Numbers are stack numbers.

AISLE	SECTION	RATING	PEAK GAMMA µSv/h	H_2 ppm	SAMPLES	WHY
A	0–2 m	RESTRICTED	1.71	< 2	1854	Drum needing action here: A-L-01-T2.
A	2–4 m	NOT CHECKED	–	–	0	The robot did not go here this time.

Samples = telemetry records logged in the section. H_2 "< n" means below the sensor's detection limit.

02 Every drum at a glance

6 DRUMS · MOST URGENT FIRST

DRUM	WHERE	RESULT	WHAT WAS SEEN	HOW SURE	FAULT VIEWS	GAMMA µSv/h
A-L-01-T2	Left 1 · middle shelf	ACTION NEEDED	Rust	High 97%	3/3	1.71
A-L-01-T3	Left 1 · top shelf	CHECK AGAIN	Nothing found	High 95%	0/6	1.35
A-L-01-T1	Left 1 · bottom shelf	NO PROBLEM FOUND	Nothing found	High 97%	0/6	1.60
A-R-01-T1	Right 1 · bottom shelf	NO PROBLEM FOUND	Nothing found	High 97%	0/3	1.23
A-R-01-T2	Right 1 · middle shelf	NO PROBLEM FOUND	Nothing found	High 96%	0/3	1.31
A-R-01-T3	Right 1 · top shelf	NO PROBLEM FOUND	Nothing found	High 96%	0/3	1.38

How sure = the detector’s confidence in what it reported: High ≥ 70%, Medium ≥ 50%, Low below. Fault views = camera views that showed damage, out of all views taken. Drum names read aisle, side (L / R), stack, shelf (T1 = bottom). This is the first report for these drums, so there is no earlier inspection to compare with.

03 Drums that need attention

2 DRUMS

A-L-01-T2 Aisle A, left side, stack 1, middle shelf

ACTION NEEDED



pass 1 · view 3 · 01:03.0 into the run · frame 1501 · body pitch 0.3°. Box and label are the detector's own output.


RUST 0.97


RUST 0.96

What was seen	Rust
How sure	High (97%)
Views	3 of 3 showed damage
Why this rating	Rust confirmed in 3 of 3 views.
Gamma at drum	Normal · 1.71 µSv/h
Air	19.1 °C · RH 62%
Hydrogen	below detection (< 2 ppm)
Surface wipe	not taken
Not measured	surface temp.
Label	read: A-L-01-T2
What the robot did	00:58 **Go to A-L-01-T2.** "The yellow barrel marked A-L-01-T2 is the next closest target to the current position and is part of the same stack as the previously inspected A-L-01-T1, making it the logical next step in the inspection sequence." 01:02 Label read as "A-L-01-T2": matches. 01:04 Rated DEFECT (Confirmed defect | verified), from frames 1428, 1453, 1501.


Label crop the reader used

> **WHAT TO DO**
>
> Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

A-L-01-T3 Aisle A, left side, stack 1, top shelf

CHECK AGAIN



retry · view 3 · 01:27.2 into the run · frame 2080 · body pitch 11.3°. Box and label are the detector's own output.



NORMAL 0.95 NORMAL 0.95 NORMAL 0.95 NORMAL 0.95 NORMAL 0.95

What was seen	Nothing found
How sure	High (95%)
Views	0 of 6 showed damage
Why this rating	The drum label could not be read after a retry, so its identity is not confirmed.
Gamma at drum	Normal · 1.35 µSv/h
Air	19.1 °C · RH 63%
Hydrogen	below detection (< 2 ppm)
Surface wipe	not taken
Not measured	surface temp.
Label	**not confirmed** · reads: AL911111111 → UNREADABLE
What the robot did	01:05 **Go to A-L-01-T3.** "The yellow barrel at A-L-01-T3 is the next target for inspection, as it is the closest and most accessible package in the current stack. The previous inspection of A-L-01-T2 confirmed a defect, so this barrel requires further examination to ensure compliance with safety standards." 01:08 Label read as "AL911111111": does not match. 01:26 Label read as "UNREADABLE": does not match. 01:28 Rated UNRESOLVED (OK | unresolved), from frames 1590, 1615, 1663, 2007, 2032, 2080.



Label crop the reader used

> **WHAT TO DO**
> Read the drum label in person and confirm which drum this is. Then repeat the inspection.

04 Drums with no problem found

4 DRUMS · BEST VIEW EACH



A-L-01-T1 97% · 6 views



A-R-01-T1 97% · 3 views



A-R-01-T2 96% · 3 views



A-R-01-T3 96% · 3 views

05 What this report cannot tell you

- **A-L-01-T3**: the drum label could not be read after a retry, so its identity is not confirmed.
- **1 section** (A 2–4 m): not walked on this round.
- **Drum surface temperature**: not measured anywhere on this round.
- **Surface wipes (swabs)**: not measured anywhere on this round.
- **Waste category (LLW / ILW)**: not measured anywhere on this round.

How accurate was the robot?

This run folder has no ground truth, so the self-check was not made. Ground truth is never written into the robot's logs; add ground_truth.json or colour_accuracy.json to score the run.

How the ratings were made

PART	DOES	NEVER DOES
Detector	Draws a box and a class (normal / rust / bulge) on each camera view.	Decide a rating.
Rules	Turn views and readings into drum results and aisle ratings, the same way every time.	Read free text.
Cosmos Reason 2	Chooses from a fixed list of options: accept the rules' result, ask for a closer look, or mark a drum for a person. Its reasons are quoted in the record.	Clear a drum, confirm damage, or change any rating, zone or number.
Report	Lays out what the run logged.	Estimate a value that was not logged.

On this round Cosmos Reason 2 made 8 reviews: 2 closer_look, 1 a-r-01-t3, 1 a-r-01-t1, 1 a-r-01-t2, 1 a-l-01-t1, 1 a-l-01-t2, 1 a-l-01-t3.

Limits used

RATING	LEVEL	SOURCE
Restricted area	gamma > 7.5 µSv/h	HSE, Work with ionising radiation (L121), regulation 17. Used as an instantaneous robot rule, not a legal controlled-area assessment.
Keep out	gamma > 100 µSv/h	HSE L121 guidance on high instantaneous dose rates.
Surface wipe	> 0.4 / 4 Bq/cm²	IAEA SSR-6 Rev. 2 (2025), para 508. Transport limits borrowed as escalation values.
Warm drum	≥ 5 °C above air	Team demo rule. Not an operational limit.
Hydrogen	no level set	No cited alarm level yet. Readings are printed; hydrogen changes no rating.

How to read the labels

ACTION NEEDED — Damage confirmed from more than one view. An engineer should assess the drum.

CHECK AGAIN — Possible damage, or the robot could not confirm the drum.

NO PROBLEM FOUND — No damage seen in this inspection.

KEEP OUT — Highest-priority simulation flag.

RESTRICTED — Raised gamma, or a damaged or unresolved drum in the section.

LOW GAMMA — Gamma measured and below 7.5 µSv/h.

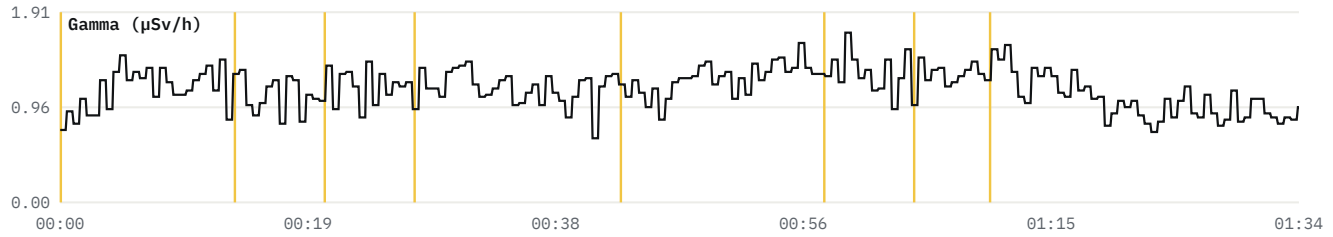
NOT CHECKED — No radiation reading here: treat as unknown.

Practice run in a computer-simulated store (NVIDIA Isaac Sim). The detector was trained on synthetic images; rust extent is judged from photos, not measured. Only a surface wipe can show contamination. The structure follows the themes of the NDA's Guidance on Monitoring of Waste Packages During Storage (WPS/640).

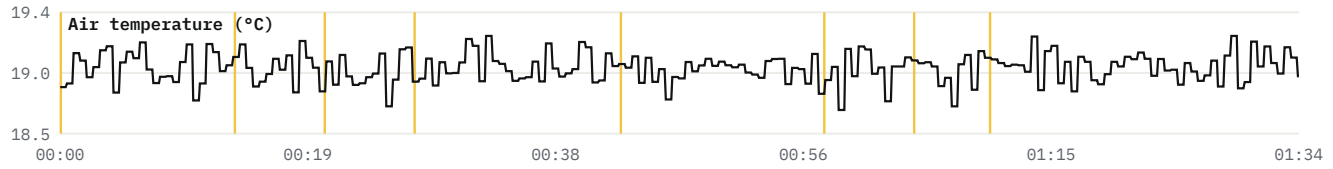
A Readings over time

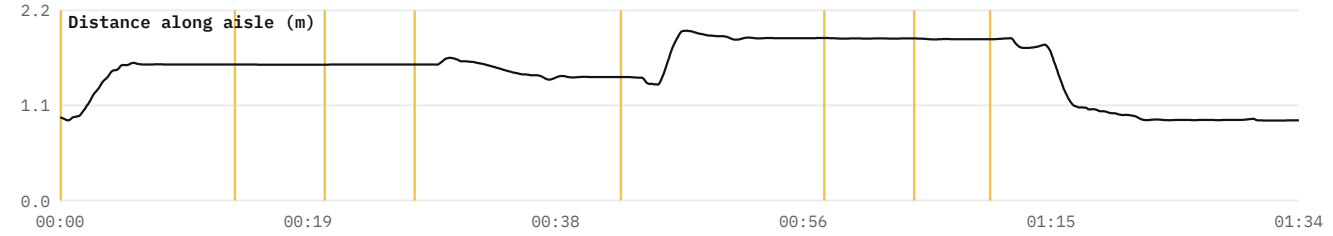
MISSION CLOCK MM:SS

Every sensor sample the robot logged. Yellow lines mark Cosmos Reason 2 decisions.



The restricted level (7.5 µSv/h) is 4.4× the highest reading, so it is off this scale.





Hydrogen: below the detection limit (2 ppm) for every logged sample.

B Step-by-step record

22 LOGGED EVENTS

What the robot decided and why, in order. Cosmos Reason 2 reasons are quoted exactly as logged (the log shortens long ones); they choose the next step and never set a rating.

TIME	DRUM	SOURCE	WHAT HAPPENED	MS
00:00.0	**A-R-01-T3**	COSMOS PLANNER	**Go to A-R-01-T3.** “The yellow barrel on the right side of the aisle is the closest and most accessible target for inspection, as it is within the robot's immediate reach and does not require crossing any obstacles or moving other packages first.”	2034
00:10.2	**A-R-01-T3**	LABEL READER	Label read as “A-R-01-T3”: matches.	
00:12.2	**A-R-01-T3**	CANIS RULES	Rated OK (OK | verified), from frames 203, 228, 276.	
00:13.2	**A-R-01-T1**	COSMOS PLANNER	**Go to A-R-01-T1.** “The yellow barrel marked A-R-01-T1 is the closest and most accessible target, located at [2.791, 2.809, 1] with clear access for inspection. It is the first in the sequence and requires immediate attention to ensure compliance with safety protocols.”	2682
00:17.0	**A-R-01-T1**	LABEL READER	Label read as “A-R-01-T1”: matches.	
00:19.0	**A-R-01-T1**	CANIS RULES	Rated OK (OK | verified), from frames 365, 390, 438.	
00:20.0	**A-R-01-T2**	COSMOS PLANNER	**Go to A-R-01-T2.** “The robot is moving to the next closest package, A-R-01-T2, which is located at [2.791, 2.809, 2]. This package is the next logical step in the inspection sequence, as it is the closest and most accessible after completing the inspection of A-R-01-T1 and A-R-01-T3. The robot will likely perform a closer look at this package to ensure it meets the required safety standards.”	3347
00:23.8	**A-R-01-T2**	LABEL READER	Label read as “A-R-01-T2”: matches.	
00:25.8	**A-R-01-T2**	CANIS RULES	Rated OK (OK | verified), from frames 527, 552, 600.	
00:26.8	**A-L-01-T1**	COSMOS PLANNER	**Go to A-L-01-T1.** “The package is positioned at the top of the stack, making it the next logical target for inspection after completing the current stack. It is also the closest package to the current position, minimizing unnecessary movement.”	1987

TIME	DRUM	SOURCE	WHAT HAPPENED	MS
00:40.4	**A-L-01-T1**	LABEL READER	Label read as “A-L-01-71”: does not match.	
00:42.5		COSMOS PLANNER	**Closer look.** “The yellow barrel marked A-L-01-T1 is partially obscured by adjacent barrels, making its label and details hard to read. A closer look is needed to confirm the label and ensure proper identification.”	1834
00:54.8	**A-L-01-T1**	LABEL READER	Label read as “A-L-01-T1”: matches.	
00:56.8	**A-L-01-T1**	CANIS RULES	Rated OK (OK | verified), from frames 924, 949, 997, 1266, 1291, 1339.	
00:57.9	**A-L-01-T2**	COSMOS PLANNER	**Go to A-L-01-T2.** “The yellow barrel marked A-L-01-T2 is the next closest target to the current position and is part of the same stack as the previously inspected A-L-01-T1, making it the logical next step in the inspection sequence.”	2152
01:01.6	**A-L-01-T2**	LABEL READER	Label read as “A-L-01-T2”: matches.	
01:03.6	**A-L-01-T2**	CANIS RULES	Rated DEFECT (Confirmed defect | verified), from frames 1428, 1453, 1501.	
01:04.7	**A-L-01-T3**	COSMOS PLANNER	**Go to A-L-01-T3.** “The yellow barrel at A-L-01-T3 is the next target for inspection, as it is the closest and most accessible package in the current stack. The previous inspection of A-L-01-T2 confirmed a defect, so this barrel requires further examination to ensure compliance with safety standards.”	2415
01:08.4	**A-L-01-T3**	LABEL READER	Label read as “AL911111111”: does not match.	
01:10.4		COSMOS PLANNER	**Closer look.** “The yellow barrel labeled "A-L-01-T3" is the target for inspection, and its label is partially obscured or unclear, requiring a closer look to verify details.”	1765
01:25.9	**A-L-01-T3**	LABEL READER	Label read as “UNREADABLE”: does not match.	
01:27.9	**A-L-01-T3**	CANIS RULES	Rated UNRESOLVED (OK | unresolved), from frames 1590, 1615, 1663, 2007, 2032, 2080.	

C Technical details

FOR ENGINEERS AND AUDITORS

Report	CANIS-112205-adaptive_aisle-20f6
Run folder	canis_wh/112205-adaptive_aisle-20f6/canis_adaptive_aisle
Input format	canis.v1
Report software	5b032b6
Simulator	Isaac Sim / Warehouse-v3 / fresh randomized physical inventory
Repository base	8bab9228b4871652760bc99605f48741f7cbb16c
Detector	Expanded YOLO26s
Detector SHA-256	ba4c80bcb99f6152d8ee0a6523151c4640ad67d727431b9ec846235dcf895b6a
Planner	nvidia/Cosmos-Reason2-8B
Planner record	Parsed model responses with original action and reason fields, plus request context
Scene seed	4382528438619680845
Camera mount	15° fixed
Surface temperature	Unresolved. The recorded LWIR patch lacks verified drum-pixel attribution; apparent temperatures are withheld from the presentation.
Log lines	4828 (336 detections, 8 planner decisions)
Status names	Action needed = Confirmed defect · Check again = Suspect or Couldn’t inspect · No problem found = OK

Evidence frames

DRUM	VIEW	TIME	FRAME	CLASS	CONF.	PITCH	SOURCE
A-L-01-T1	pass 1 · view 1	00:38.3	924	normal	0.965	-9.8°	2ed77e0ece9863ea…
A-L-01-T1	pass 1 · view 2	00:39.8	949	normal	0.967	-9.3°	2c64e164b575a34e…
A-L-01-T1	pass 1 · view 3	00:41.8	997	normal	0.965	-8.4°	87753a81207ee2a3…
A-L-01-T1	retry · view 1	00:52.9	1266	normal	0.966	-7.9°	67413db2f2cf70a9…
A-L-01-T1	retry · view 2	00:53.9	1291	normal	0.965	-7.2°	46e4347805a9de11…
A-L-01-T1	retry · view 3	00:55.9	1339	normal	0.964	-6.3°	0abdef109e4ce061…
A-L-01-T2	pass 1 · view 1	00:59.5	1428	rust	0.966	-1.3°	248c41aeea52fbde…
A-L-01-T2	pass 1 · view 2	01:01.0	1453	rust	0.964	-0.5°	42af345547556f39…
A-L-01-T2	pass 1 · view 3	01:03.0	1501	rust	0.967	0.3°	a8edaf23efceae23…
A-L-01-T3	pass 1 · view 1	01:06.5	1590	normal	0.949	9.6°	c9b074ad52fec2fa…
A-L-01-T3	pass 1 · view 2	01:07.5	1615	normal	0.951	10.4°	4904aaa69556e6f1…
A-L-01-T3	pass 1 · view 3	01:09.6	1663	normal	0.951	11.2°	7435f062ffa2fe5a…
A-L-01-T3	retry · view 1	01:24.2	2007	normal	0.953	9.8°	f7c50e4b433025fa…
A-L-01-T3	retry · view 2	01:25.2	2032	normal	0.951	10.5°	f17188cd58390280…
A-L-01-T3	retry · view 3	01:27.2	2080	normal	0.954	11.3°	77fe64168f0f0762…
A-R-01-T1	pass 1 · view 1	00:15.1	365	normal	0.967	-10.0°	560cbab43ce47733…
A-R-01-T1	pass 1 · view 2	00:16.1	390	normal	0.969	-9.2°	11ca352d103b5059…
A-R-01-T1	pass 1 · view 3	00:18.1	438	normal	0.968	-8.4°	3edb13bab3cae134…
A-R-01-T2	pass 1 · view 1	00:21.7	527	normal	0.959	-1.3°	094ba66ba89ef47c…

DRUM	VIEW	TIME	FRAME	CLASS	CONF.	PITCH	SOURCE
A-R-01-T2	pass 1 · view 2	00:22.7	552	normal	0.960	-0.5°	9f390fcc0646f629…
A-R-01-T2	pass 1 · view 3	00:24.7	600	normal	0.960	0.4°	c66ffe84b4429098…
A-R-01-T3	pass 1 · view 1	00:08.1	203	normal	0.952	9.7°	e812cdb3061a5377…
A-R-01-T3	pass 1 · view 2	00:09.1	228	normal	0.952	10.4°	6e19b36195ddab99…
A-R-01-T3	pass 1 · view 3	00:11.1	276	normal	0.956	11.3°	ea4f6de189c76d76…